Case 17: Spatial convergence

1 Verification purpose

This isolated-process case verifies spatial convergence of ELMFIRE's level-set propagation for the same manufactured circular-front problem on four meshes. It exercises raster-derived grid geometry, front initialization, propagation, and level-set field output while holding physical inputs and temporal controls fixed. It can reveal grid-spacing misuse, nonconvergent front updates, or resolution-dependent physical configuration. The question is whether all four required meshes complete, the fitted radius-error order is at least 0.8, and the finest-grid error is at most 1 m.

2 Mathematical formulation and numerical implementation

For $\partial_t \phi + R_0 |\nabla \phi| = 0$ with constant $R_0 = 0.0116332$ m s^{-1}, the exact circular radius is mesh-independent. ELMFIRE obtains Δx from each GeoTIFF transform. The four simulation conditions use 128, 256, 512, and 1024 cells over the same 256 m domain, so $\Delta x = \{2, 1, 0.5, 0.25\}$ m.

3 Derivation of expected behavior

With $r_0 = 20$ m and $T = 360$ s, $r_\star = 20 + R_0 T = 24.187952$ m. For a consistent spatial discretization, the zero-contour equivalent-radius error $E_h = |r_A - r_\star|$ should decrease as $h = \Delta x$ decreases and follow $E_h \approx C h^p$. A least-squares slope of $\log E_h$ against $\log h$ estimates p.

Every grid simulation condition uses the same manufactured solution. From the generated rate $R_s = 0.0116332$ m s^{-1}, initial radius $r_0 = 20$ m, and stop time $T = 360$ s,

$$r_{\text{ref}} = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For each cell size $h_i \in \{2, 1, 0.5, 0.25\}$ m, the area-equivalent radius and error are

$$r_i = \sqrt{\frac{N_{b,i} h_i^2}{\pi}}, \qquad E_i = |r_i - 24.187952|.$$

The observed spatial order is the ordinary-least-squares slope

$$p_h = \frac{\sum_i (\ln h_i - \overline{\ln h})(\ln E_i - \overline{\ln E})}{\sum_i (\ln h_i - \overline{\ln h})^2}.$$

Consequently the table tests the explicitly calculated $p_h \geq 0.8$ and the finest-grid substitution $E_{0.25} = |r_{0.25} - 24.187952| \leq 1$ m.

4 Verification metrics and acceptance criteria

The first metric is the fitted order p, selected because it measures the multi-grid trend rather than one favorable mesh; $p \geq 0.8$ allows rasterized-interface effects while requiring approximately first-order behavior. The second is finest-grid $E_h \leq 1$ m, preventing a misleading order from large absolute errors. Radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, providing a rotationally aggregated subcell measure. The last matching level-set field is required for each simulation condition. All simulation conditions and both metrics must pass; otherwise the decision is NOT EVALUABLE or FAIL as appropriate.

5 Simulation configuration and predicted outcome

Each EPSG:32610 simulation condition covers the same 256 m square with uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, $R_0 = 0.0116332\,\mathrm{m\,s^{-1}}$, and duration 360 s. Only grid resolution varies. The predicted front is circular and E_h decreases toward the 24.187952 m reference radius.

Parameter	Configured value	Units	Role
Domain	256 by 256	m	Fixed physical experiment
Meshes	128, 256, 512, 1024	cells/side	Spatial refinement sequence
Cell size	2, 1, 0.5, 0.25	m	Independent variable
Time control	step 1; stop 360	s	Fixed temporal discretization
Initial front	centered signed-distance radius 20	m	Common exact geometry

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

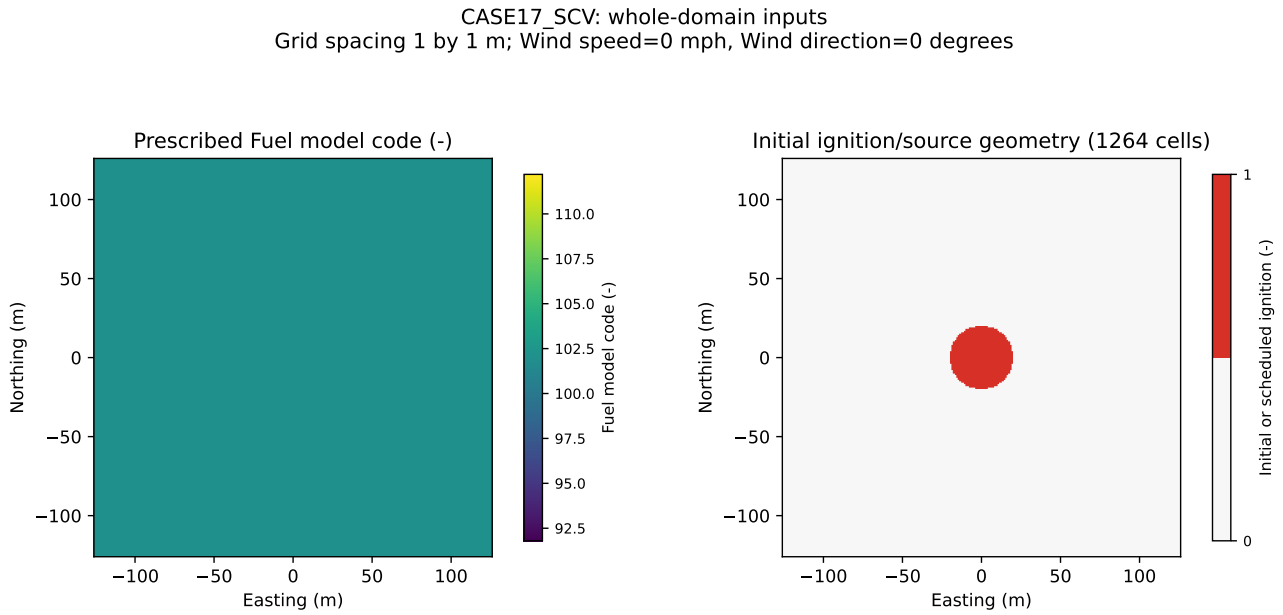


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only the four case-specific simulation condition outputs are read. Current status is **FAIL**; 4 simulation conditions and 2 decision metrics are recorded.

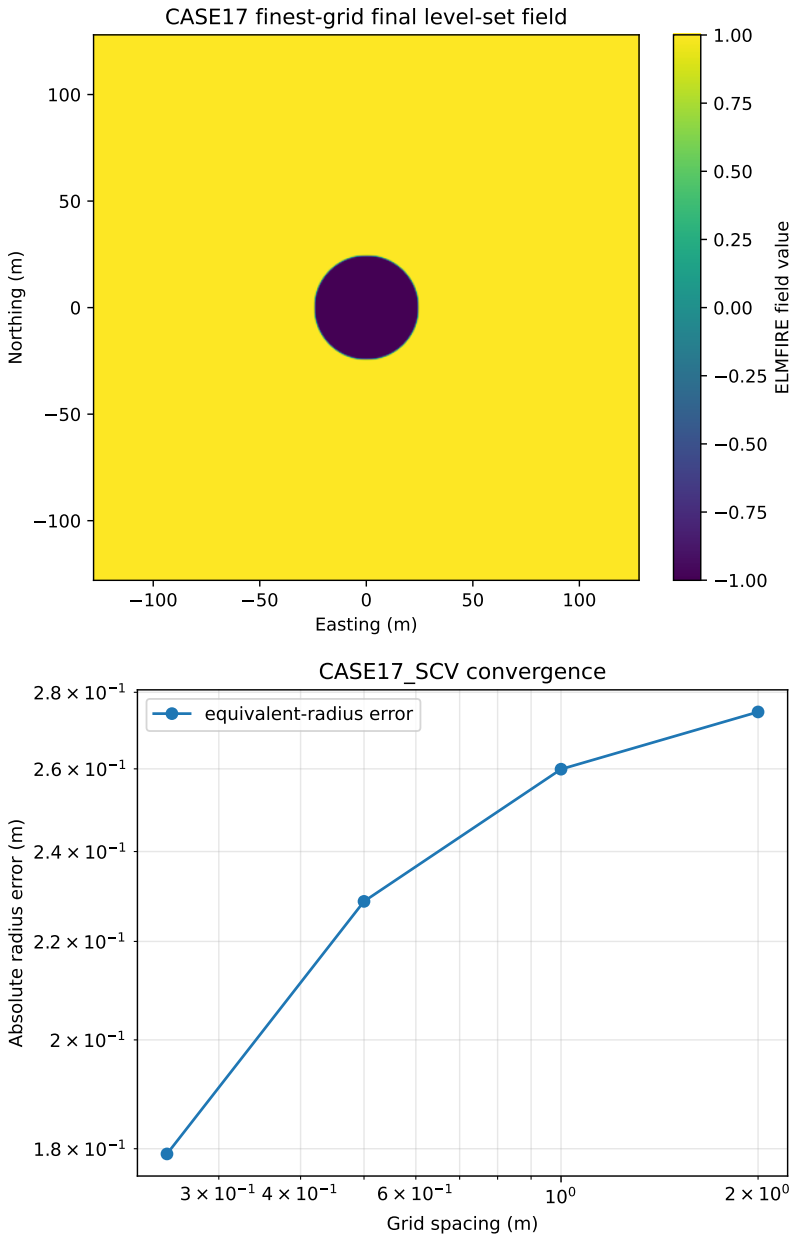


Figure 2: Finest-grid whole-domain signed level-set result and radius-error convergence across all four completed meshes. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed spatial order	0.8 ($\geq$ threshold)	0.203637 -	FAIL
finest-grid radius error	0 ($\leq$ 1)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

Low order with small finest error suggests interface-measurement or pre-asymptotic effects; high order with excessive finest error indicates the sequence has not reached useful accuracy. Missing simulation conditions invalidate the convergence fit. Resolution-dependent physical inputs would be a configuration defect.

9 Scope and limitations

This case measures convergence of one smooth, radially symmetric front and an area-derived observable. It does not determine formal order for discontinuous fuels, complex terrain, anisotropic spread, or other output fields.